

Online Resource 1. ODD Protocol and Model Documentation for the Religious Environmental Mobilization Model

Purpose

The purpose of the model is to examine the conditions under which religious environmental belief remains symbolic and the conditions under which it becomes collective environmental mobilization. The model is designed for theoretical mechanism clarification rather than empirical prediction. It formalizes a belief-to-action process in which stewardship belief, climate risk perception, trust in science, religious leadership, community norms, institutional capacity, political filtering, activism costs, and participation thresholds interact to generate symbolic environmentalism, concern, norm support, mobilization, or resistance.

The model asks why religious environmental concern does not automatically become collective action. It treats religious environmental mobilization as an emergent outcome of interacting individual, communal, institutional, and political mechanisms. The central theoretical assumption is that stewardship belief may increase environmental concern, but mobilization requires additional enabling conditions, including supportive leadership, strong community norms, institutional capacity, manageable activism costs, and sufficiently low participation thresholds.

Entities, State Variables, and Scales

The model contains 1,000 agents distributed across 20 religious communities. Each agent represents an individual member of a religious community. Agents are not intended to represent empirically observed individuals. They are stylized theoretical actors used to examine how different mechanisms interact over simulated time.

Each agent has the following state variables:

State variable	Description
community identity	The religious community to which the agent belongs.
leader type	The environmental framing orientation of the agent's community leader.
stewardship belief	The degree to which the agent accepts environmental care as a religious or moral responsibility.
trust in science	The agent's openness to scientific claims about climate risk.
trust in religious leaders	The agent's tendency to accept religious leadership as morally authoritative.
political identity	The agent's political orientation, including possible resistance to environmental mobilization under conditions of polarization.
climate risk exposure	The agent's perceived exposure to climate risk.
community norm strength	The perceived strength of pro-environmental norms within the agent's religious community.
perceived cost of activism	The social, emotional, political, or practical cost attached to joining collective environmental action.

State variable	Description
environmental concern	The agent's level of concern about environmental or climate issues.
moral obligation	The agent's sense that environmental action is morally or religiously required.
mobilization score	The agent's propensity to join collective environmental action.
mobilization status	The agent's current outcome category.

All main attitudinal and mechanism variables are bounded between 0 and 100. Religious communities are assigned one of four leader types: silent, moderate, activist, or skeptical. Leader types differ in their environmental framing values. A silent leader provides little or no environmental framing. A moderate leader provides limited positive framing. An activist leader strongly supports environmental action. A skeptical leader discourages or undermines environmental mobilization.

Agents can occupy one of six statuses: passive, concerned, symbolic steward, norm supporter, mobilized, or resistant. Passive agents remain inactive. Concerned agents have environmental concern but do not translate it into moral obligation or action. Symbolic stewards adopt environmental care as a moral or religious value without mobilizing. Norm supporters support pro-environmental community norms but do not join collective action. Mobilized agents join collective environmental action. Resistant agents reject or resist environmental mobilization.

Process Overview and Scheduling

At initialization, agents are assigned to religious communities and receive heterogeneous values around scenario-specific baselines. At each tick, agents update their climate risk exposure, community norm strength, environmental concern, moral obligation, mobilization score, and mobilization status.

Climate risk exposure gradually moves toward the scenario-level climate risk parameter. Community norm strength is calculated from the share of norm supporters and mobilized agents in the agent's community, combined with the leader's environmental framing value. Environmental concern, moral obligation, and mobilization score are updated using weighted equations. Agents are then classified into statuses through a hierarchical decision rule.

The model proceeds in the following sequence at each tick:

1. Agents update perceived climate risk exposure.
2. Community norm strength is recalculated within each religious community.
3. Agents update environmental concern.
4. Agents update moral obligation.
5. Agents update mobilization score.
6. Agents are assigned to one of six mobilization statuses.

Each scenario is run for 150 ticks and repeated 30 times. Results are reported as mean values across repetitions. The repeated runs are used to reduce the effect of random initialization and stochastic fluctuations.

Design Concepts

The model includes heterogeneity, interaction, feedback, thresholds, and stochasticity.

Agents are heterogeneous in stewardship belief, trust in science, trust in religious leaders, political identity, climate risk perception, and cost sensitivity. This heterogeneity allows the model to represent variation within religious communities rather than assuming uniform religious responses to environmental concern.

Interaction occurs through community-level norm formation. Agents are influenced by the proportion of norm supporters and mobilized agents in their own religious community. This means that mobilization is not only an individual-level decision, but also a community-level process shaped by visible normative environments.

Feedback occurs when norm supporters and mobilized agents increase community norm strength. Stronger community norms can then increase later moral obligation and mobilization among other agents. This creates the possibility of cumulative activation, where early norm support or mobilization alters the conditions for later mobilization.

Thresholds determine whether an agent becomes mobilized. The mobilization threshold represents the barrier between belief and action. Even when agents have environmental concern and moral obligation, they may not mobilize if the mobilization score remains below the threshold. The threshold therefore captures the social, emotional, political, institutional, and practical barriers that separate environmental belief from collective action.

Stochasticity is introduced through random initialization and small fluctuations in perceived climate risk. The purpose of stochasticity is not to reproduce empirical randomness, but to prevent the model from producing purely deterministic outcomes based only on fixed parameter values.

Initialization

The baseline model initializes 1,000 agents across 20 religious communities. Baseline values are set for stewardship belief, trust in science, political identity, climate risk level, institutional capacity, activism cost, and mobilization threshold. Agents receive individual values through bounded random distributions around these baselines.

The baseline leader distribution contains silent, moderate, activist, and skeptical leaders. Scenario-specific conditions modify one or more parameters while keeping the model structure constant. For example, some scenarios increase stewardship belief, some increase climate risk exposure, some strengthen activist leadership, some increase institutional capacity, and some combine multiple enabling conditions.

The model is intentionally abstract. It does not represent a specific religious tradition, country, organization, or environmental movement. Its variables are stylized theoretical constructs rather than empirically calibrated measures. This abstraction is appropriate because the purpose of the model is mechanism clarification rather than prediction.

Input Data

The model does not use empirical microdata or survey-based calibration. The selection of variables is theory-driven and literature-informed. The numerical weights and parameter values are stylized theoretical assumptions used to explore plausible directional relationships.

No human-subject data are used. The model is not calibrated to a specific religious population, denomination, country, organization, survey, or movement. The results should therefore be interpreted as theoretical mechanism clarification rather than empirical prediction.

Submodels

The model includes five main submodels: climate risk updating, community norm updating, environmental concern, moral obligation, and mobilization score. A final status classification rule assigns agents to outcome categories.

Climate Risk Updating

Climate risk exposure updates gradually toward the scenario-level climate risk parameter. This means that agents do not immediately adopt the full scenario-level risk value. Instead, perceived risk changes over time. Small stochastic fluctuations are included to represent minor variation in perceived exposure.

Climate risk exposure contributes to environmental concern. Higher perceived climate risk generally increases concern, especially among agents with higher trust in science.

Community Norm Updating

Community norm strength is calculated from the share of norm supporters and mobilized agents in the agent's religious community, combined with the leader's environmental framing value. This submodel represents the idea that environmental action becomes more plausible when it is socially visible and normatively supported within the religious community.

Leader type matters because religious leaders shape the moral framing of environmental concern. Activist leaders increase the visibility and legitimacy of environmental action. Moderate leaders provide weaker but still positive support. Silent leaders do not strongly influence mobilization. Skeptical leaders reduce or block the development of pro-environmental community norms.

Environmental Concern

Environmental concern is shaped by climate risk exposure, trust in science, stewardship belief, and political identity. Agents with higher climate risk exposure and higher trust in science are more likely to develop environmental concern. Stewardship belief can also increase concern by framing environmental care as religiously or morally meaningful.

Political identity can filter or reduce concern, especially under conditions of polarization. This reflects the assumption that climate concern may be weakened when environmental issues are interpreted through politicized or oppositional frames.

Moral Obligation

Moral obligation is shaped by stewardship belief, trust in religious leaders, community norms, and leader framing. This submodel distinguishes environmental concern from moral or religious obligation. An agent may be concerned about climate change without experiencing environmental action as a religious duty.

Moral obligation increases when stewardship belief is high, when religious leaders provide supportive environmental framing, when agents trust religious leaders, and when pro-environmental norms are visible within the community.

Mobilization Score

Mobilization score is shaped by environmental concern, moral obligation, community norm strength, institutional capacity, activism cost, and political identity. Concern and moral obligation increase the propensity for action, but they are not sufficient on their own. Community norms and institutional capacity help convert concern into action by making participation more socially supported and practically available.

Activism cost reduces mobilization. Political filtering can also reduce mobilization when environmental action is perceived as politically threatening, socially risky, or ideologically incompatible. Institutional capacity increases mobilization by providing organizational pathways, resources, and opportunities for participation.

Status Classification

After the mobilization score is calculated, agents are assigned to one of six statuses through a hierarchical decision rule.

Agents become mobilized when their mobilization score exceeds the mobilization threshold. Agents become norm supporters when they support pro-environmental community norms but do not cross the mobilization threshold. Agents become symbolic stewards when they hold stewardship belief and moral obligation but do not mobilize. Agents become concerned when they have environmental concern but lack sufficient moral obligation or mobilization propensity. Agents become resistant when political filtering, skeptical leadership, or low trust conditions produce opposition to environmental mobilization. Agents who do not meet these conditions remain passive.

This classification system allows the model to distinguish between different non-mobilized outcomes. Non-mobilization is not treated as a single category. Agents may remain passive, become concerned, adopt symbolic stewardship, support norms, resist mobilization, or join collective action.

Simulation Experiments

The model is used to compare theoretically meaningful scenarios. Each scenario modifies one or more parameters while holding the basic model structure constant. The main scenarios include baseline, high stewardship, high risk, activist leader, skeptical leader, high polarization, high capacity, risk plus activist leadership, activist leadership plus capacity, risk plus capacity, combined activation, and full alignment.

The purpose of the scenarios is to examine whether stewardship belief alone is sufficient for mobilization, or whether mobilization requires alignment among concern, moral obligation, community norms, leadership, institutional capacity, and low participation barriers.

A sensitivity analysis is also conducted on the mobilization threshold. The main simulations use a threshold of 40. Additional simulations use threshold values of 30, 35, 40, 45, and 50. This test examines whether the findings depend on the chosen threshold value and whether the threshold itself operates as a substantive mechanism in the belief-to-action process.

Model Scope and Limitations

The model is intentionally theoretical and generative. It does not predict real-world levels of religious environmental mobilization. It does not represent a specific religious community, national context, denomination, or environmental organization. Its purpose is to clarify mechanisms that may explain why religious environmental concern sometimes remains symbolic and sometimes becomes collective environmental action.

The model also simplifies religious communities by treating them as bounded communities with identifiable leader types and shared norm environments. In empirical cases, religious communities are more internally heterogeneous, networked, and contested. Political identity, trust in science, religious authority, and institutional capacity are also more complex than single numerical variables can capture.

Future versions of the model could introduce social networks, dynamic leadership, organizational change over time, multiple forms of environmental action, movement-counter movement dynamics, media effects, inter-community diffusion, and empirically calibrated parameter distributions.

Despite these limitations, the model is useful for formalizing the belief-to-action transition. By separating stewardship belief, environmental concern, moral obligation, community norms, institutional capacity, political filtering, activism costs, and mobilization thresholds, it allows religious environmental mobilization to be examined as a process rather than assumed as a direct outcome of religious environmental belief.